Project 1 — Edition A

Company: Microsoft Corporation (MSFT)
Valuation date: September 1, 2026
Decision and user: Equity-research recommendation to a buy-side investment committee with no current Microsoft position.

Initial decision: watch / defer

I recommend **watch / defer** rather than initiate a position at this stage. Microsoft has compelling operating momentum and a strong liquidity position, which justify deeper valuation work. They do not, by themselves, demonstrate undervaluation. An initiation decision should wait until a defensible intrinsic-value range, a dated market-price comparison, and downside sensitivity analysis are complete.

Known evidence from the current primary filing

Primary source: Microsoft Corporation Form 10-K, accession 0001193125-26-323660; reporting period ended June 30, 2026; filed July 29, 2026.

Item	Filing evidence
Entity	Microsoft Corporation; CIK 0000789019; ticker MSFT
Currency / units	U.S. dollars; cited financial-statement amounts are in millions except per-share amounts
Diluted-share evidence	7,453 million FY2026 diluted weighted-average shares: 7,429 million basic weighted-average shares plus 24 million dilutive stock-based awards. This is distinct from the 7,425,545,491 shares outstanding reported on the cover as of July 23, 2026.
Cash	$20,935 million cash and cash equivalents plus $55,908 million short-term investments = $76,843 million
Debt	$9,227 million current portion of long-term debt plus $31,067 million long-term debt = $40,294 million
Operating-value driver	Microsoft Cloud revenue increased 27% to $214.4 billion; Azure and other cloud-services revenue increased 41%.

The Week 1 screening case also listed MSFT as an eligible operating company with 14% annualized return, 24% annualized volatility, 54% positive-day share, no missing observations, and a filing available. These figures are expressly synthetic training data, not market evidence, so they support only the provisional decision to investigate. Week 1 screening case

MD&A reading notes

Source locator for all notes below: FY2026 10-K, Item 7, MD&A, “Overview” and “Fiscal Year 2026 Compared with Fiscal Year 2025,” pp. 33–44; Item 1 and Item 1A for the business description and risk

factors. 10-K

- **What Microsoft sells:** Management describes a mixed software-and-cloud business: cloud services, software licensing and support, online advertising, and devices/gaming. My note: the key economic shift is toward recurring and consumption-based cloud revenue, but Microsoft still has exposed PC, gaming, advertising, and on-premises revenue streams.
- **What changed and why:** Revenue increased $50.1 billion (18%), which management attributes principally to Microsoft Cloud growth, especially Azure and Microsoft 365 Commercial cloud. My note: the growth story is broad enough to merit deeper work, but Azure is the load-bearing driver; the valuation should not simply extend the consolidated 18% rate.
- **The margin and investment story:** Cost of revenue rose 21% and Microsoft Cloud gross margin declined to 66%. Management attributes this to AI infrastructure investment and higher AI product usage, partly offset by efficiency gains. My note: AI is not just a demand catalyst—it is currently a cost and capital-intensity question. The model must test whether incremental cloud revenue produces enough margin and free cash flow to justify the infrastructure buildout.
- **Management's warning signals:** The filing identifies intense technology competition, cybersecurity and privacy threats, global/economic and geopolitical exposure, and the need to adapt product categories, business models, and sales motions. My note: the thesis fails if cloud growth depends on permanently elevated infrastructure spending, if competition compresses cloud economics, or if a security/regulatory event materially harms customer trust.

MD&A follow-up checks

1. Reconcile Azure growth, cloud gross margin, capital expenditures, and depreciation over multiple filings to identify whether the infrastructure burden is temporary or structural.
2. Assess commercial remaining performance obligation ($678 billion, up 84%) for duration, cancellation, and revenue-recognition timing; it is not equivalent to current revenue.
3. Build separate revenue and margin cases for Azure/cloud, Microsoft 365, and the smaller consumer/device/gaming businesses; avoid a single-growth-rate model.

Synthetic enterprise-to-equity bridge check

All amounts are USD millions except per-share values.

Step	Calculation	Result
Enterprise value	Given	$1,000
Add nonoperating cash	$1,000 + $100	$1,100
Less debt	$1,100 – $250	$850
Less noncontrolling interest	$850 – $20	$830
Less unfunded pension	$830 – $10	$820 equity value
Divide by diluted shares	$820 / 100 million	**$8.20 per share**

The bridge demonstrates that enterprise value becomes common-equity value only after nonoperating cash is added and debt plus other senior/non-common claims are removed; the result is then divided by diluted shares. This is a classroom mechanics check, not a Microsoft valuation.

Consequential assumptions

- FCFF and enterprise value are appropriate for Microsoft as an operating company.
- Azure/cloud growth should moderate over a finite forecast horizon; the most recent growth rate is not a perpetual-growth assumption.
- AI infrastructure spending may pressure margins and capital intensity before it supports free-cash-flow growth.
- The final bridge must apply a consistent policy to cash, short-term investments, debt, leases, pensions, noncontrolling interests, and other nonoperating items.
- Diluted weighted-average shares are the appropriate working denominator, subject to the final valuation-date convention.

Unknowns and research plan

Key unknowns are the market price and implied multiples as of the valuation date; the durability of cloud demand; AI data-center capital requirements; operating-margin and cash-conversion paths; WACC; terminal growth; and the appropriate treatment of nonoperating assets and claims.

I will:

1. Reconcile a three-statement historical dataset to Microsoft’s 10-Ks and 10-Qs.
2. Forecast revenue by major driver, including Azure/cloud, Microsoft 365, LinkedIn, Dynamics, Windows/devices, gaming, and search.
3. Derive unlevered FCFF from EBIT, taxes, depreciation and amortization, capital expenditures, and operating working-capital investment.
4. Estimate a documented WACC and test multiple reasonable discount-rate and terminal-growth assumptions.
5. Use DCF and comparable-company cross-checks to establish an intrinsic-value range and a dated enterprise-to-equity bridge.
6. Test downside cases for slower cloud growth, lower margins, heavier AI infrastructure investment, and a higher discount rate. An initiation recommendation requires a margin of safety that survives a reasonable downside case.

Falsification question

Partner name: Connor guthrie
Partner-generated question: What evidence would convince you that Microsoft's investments in AI infrastructure are generating sufficient free cash flow to justify its current valuation, and would that change your watch/defer recommendation?

Why it matters: This question challenges my initial recommendation by focusing on the relationship between Microsoft's capital expenditures, cloud revenue growth, and free-cash-flow generation. If Microsoft demonstrates that its AI infrastructure investments produce sustainable revenue growth, improving margins, and stronger free-cash-flow conversion, my current watch/defer recommendation could be reconsidered. Conversely, if capital expenditures continue to rise without corresponding improvements in cash flow, the investment thesis would remain cautious.

Submission record

- Edition A checkpoint receipt / attempt identifier: brightspace quizzes
- Receipt timestamp: September 1, 2026, at 2:35 PM
- Submitted file: Project_1_Edition_A_MSFT.md (Brightspace)
- Version status: Original Edition A preserved for comparison with Edition B.
- Preserve this exact submitted file unchanged after receipt verification.

Project 1 — Edition B: Microsoft Equity Research

Company: Microsoft Corporation (NASDAQ: MSFT)
Valuation date: September 1, 2026
Analyst role: Buy-side equity research analyst
Investment committee position: No existing Microsoft position
Recommendation: WATCH / DEFER
Primary financial source: Microsoft FY2026 Form 10-K, fiscal year ended June 30, 2026
Valuation currency: USD; financial statement amounts in millions unless otherwise indicated

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1. Executive Summary and Investment Recommendation

I recommend **watch/defer** rather than initiating a position in Microsoft Corporation at the September 1, 2026 reference share price of $501.02.

Microsoft is a high-quality technology company with strong positions in enterprise software, cloud infrastructure, productivity applications, and artificial intelligence. Its recurring revenue streams, enterprise customer relationships, and expanding cloud business provide meaningful long-term growth opportunities.

However, business quality does not automatically establish investment attractiveness at a particular market price.

My updated valuation analysis produces three reference estimates:

Valuation methodology	Estimated value per share
FCFE pro-forma valuation	$116.94
FCFF discounted cash flow valuation	$197.71
Comparable-company median P/E valuation	$371.30
Reference market price	$501.02

All three model estimates fall below the reference share price.

The intrinsic valuation models are particularly sensitive to Microsoft's future free-cash-flow generation, AI infrastructure investment, and long-term growth assumptions. The comparable-company valuation produces a higher estimate but remains below the reference price and is sensitive to peer selection.

My recommendation is therefore to continue monitoring Microsoft rather than initiate a position without a stronger margin of safety.

Model status: The FCFF DCF, FCFE pro-forma, and comparable-company calculations are reproducible using the Python scripts in the project repository. Microsoft's FY2026 operating cash flow, capital expenditures, and September 1, 2026 reference closing price have been checked against their respective sources. The comparable-company price and EPS conventions require further verification, and the FCFF and FCFE models remain methodologically distinct rather than fully reconciled.

2. Evolution of the Investment Thesis From Edition A

Edition A established an initial **watch/defer** recommendation.

At that stage, I identified Microsoft as a financially strong business with compelling cloud growth but concluded that additional valuation work was necessary before recommending an investment.

The initial research emphasized:

- Strong Azure and Microsoft Cloud growth.
- Microsoft's established enterprise software ecosystem.
- Significant spending on AI infrastructure.
- Uncertainty about future margins and free-cash-flow conversion.
- The need to distinguish business quality from valuation attractiveness.

Edition B incorporates quantitative analysis completed during the course.

Research Progression

Area	Edition A	Edition B
Investment thesis	Initial watch/defer	Watch/defer supported by valuation analysis
Financial research	FY2026 10-K review	Historical input verification and forecasting
FCFF DCF	Planned	Completed; $197.71/share
Three-statement model	Planned	Simplified FY2027–2031 pro-forma completed
FCFE valuation	Not available	$116.94/share
Comparable-company analysis	Planned	Oracle and Adobe P/E comparison
Sensitivity analysis	Planned	Discount-rate, terminal-growth, revenue-growth, and margin scenarios
Model validation	Planned	Accounting and scenario checks
Interactive model	Not available	Streamlit valuation workbench

The original recommendation has not changed, but the supporting evidence has become more quantitative and testable.

3. Company Overview and Business Fundamentals

Microsoft operates a diversified technology business with activities in productivity software, enterprise cloud infrastructure, business applications, personal computing, gaming, and digital advertising.

The company benefits from established enterprise relationships, subscription-based revenue, and a broad portfolio of products and services.

Its major growth opportunities include Azure cloud infrastructure, Microsoft 365, Dynamics, security products, and AI-enabled software and services.

FY2026 Operating Evidence

According to Microsoft's FY2026 Form 10-K:

- Microsoft Cloud revenue increased 27% to approximately $214.4 billion.
- Azure and other cloud services revenue increased 41%.
- Consolidated revenue increased approximately 18%.
- Microsoft Cloud gross margin declined to approximately 66%.

These figures illustrate strong demand for Microsoft's cloud services alongside meaningful infrastructure-related cost pressure.

The central question is whether cloud and AI revenue growth will ultimately translate into sufficient incremental free cash flow to justify the capital invested.

Financial Strength

The FY2026 balance sheet figures used in the analysis include:

Financial item	Amount ($m)
Revenue	331,839
Total assets	758,376
Total liabilities	315,989
Stockholders' equity	442,387
Cash and cash equivalents	20,935
Short-term investments	55,908
Current and long-term debt	40,294

Cash and short-term investments total $76,843 million, compared with $40,294 million of current and long-term debt.

This provides Microsoft with financial flexibility, although liquidity does not eliminate the risks associated with large infrastructure investments.

4. Industry Position and Competitive Risks

Microsoft benefits from strong competitive positions in enterprise software and cloud computing.

Its integrated product ecosystem creates opportunities to distribute AI capabilities through existing customer relationships and software subscriptions.

However, several risks remain relevant.

Cloud competition: Microsoft competes with other major cloud infrastructure and software providers. Competition may influence pricing, customer acquisition, and infrastructure economics.

AI capital intensity: AI services require substantial investment in data centers, computing infrastructure, and related assets.

Margin pressure: Higher depreciation, infrastructure operating costs, and changing product mix could reduce margins.

Cybersecurity and regulatory exposure: Security incidents, privacy issues, and regulatory restrictions could affect Microsoft's costs, operations, or customer relationships.

Forecast uncertainty: Rapid technological change makes it difficult to predict sustainable long-term growth and returns on invested capital.

These risks are incorporated into the valuation analysis through forecast assumptions, sensitivity tests, and a cautious investment recommendation.

5. FCFF Discounted Cash Flow Valuation

The FCFF model estimates Microsoft's operating enterprise value by forecasting free cash flow to the firm and discounting those cash flows using the weighted average cost of capital.

A terminal value captures cash flows beyond the explicit five-year forecast period.

5.1 Model Assumptions

Assumption	Base-case input
Starting FY2026 FCFF	$69,446.2199 million
FY2027 FCFF growth	15%
FY2028 FCFF growth	13%
FY2029 FCFF growth	11%
FY2030 FCFF growth	9%
FY2031 FCFF growth	7%
WACC	10%
Terminal growth rate	3%
Cash and short-term investments adjustment	$76,843 million
Debt adjustment	$40,294 million
Diluted share denominator	7,453 million

The declining growth path reflects an assumption that Microsoft's free-cash-flow growth moderates over time.

The 10% WACC and 3% terminal growth rate are analyst assumptions, not directly reported financial statement figures. Their reasonableness must be evaluated through sensitivity analysis.

The model uses FY2026 diluted weighted-average shares as its working denominator. This differs from actual shares outstanding at the valuation date and represents a simplifying convention.

5.2 Starting Free Cash Flow

The provisional starting FCFF is calculated as:

FCFF = Operating cash flow + After-tax interest adjustment – Capital expenditures

Using the working inputs:

FCFF = 182,935 + 2,459.2199 – 115,948

Starting FCFF = $69,446.2199 million

The FY2026 operating cash flow of $182,935 million and capital expenditures of $115,948 million were checked against Microsoft's FY2026 Form 10-K. The after-tax interest adjustment of $2,459.2199 million was calculated using interest expense of $3,051 million and an effective tax rate derived from reported income before taxes and income tax expense. These inputs support the revised starting FCFF of $69,446.2199 million.

5.3 Five-Year FCFF Forecast

Fiscal year	Forecast FCFF ($m)
FY2027	79,863.2
FY2028	90,245.4
FY2029	100,172.4
FY2030	109,187.9
FY2031	116,831.0

5.4 Enterprise-to-Equity Valuation Bridge

Valuation component	Amount ($m)
Present value of explicit FCFF	369,566.4
Present value of terminal value	1,067,416.5
Enterprise value	1,436,982.9
Add: cash and short-term investments	76,843.0
Less: debt	(40,294.0)
Equity value	1,473,531.9
Diluted shares (millions)	7,453

Valuation component	Amount ($m)
Value per diluted share	**$197.71**

The terminal value represents approximately 74% of enterprise value.

This creates substantial sensitivity to long-term assumptions, particularly WACC and terminal growth.

The bridge is simplified and does not separately adjust for every possible nonoperating asset or non-common claim. The classification of cash, leases, pensions, and other liabilities requires consistent treatment before the model is considered fully reconciled.

6. FCFE Pro-Forma Financial Model

A separate financial model was developed to forecast Microsoft's financial statements and estimate free cash flow to equity.

The model projects FY2027 through FY2031 and incorporates revenue growth, profitability, depreciation, capital expenditures, and selected balance-sheet assumptions.

6.1 Key Operating Assumptions

Assumption	Base case
FY2027–FY2031 revenue growth	12%, 11%, 10%, 9%, 8%
Gross margin	68%
Operating expenses / revenue	21.2%
Depreciation / opening PP&E	12.3%
Tax rate	19.4%
Inventory days	5
Cost of equity	10%
Terminal growth	3%

6.2 Capital Expenditure Forecast

Microsoft's cloud and AI infrastructure investments are central to the financial forecast.

Fiscal year	Forecast capital expenditures ($m)
FY2027	120,000
FY2028	125,000
FY2029	130,000
FY2030	135,000
FY2031	140,000

These amounts are analyst forecast assumptions rather than management guidance.

The elevated investment path reflects the possibility that Microsoft's AI and cloud expansion requires sustained infrastructure spending.

6.3 Revenue and FCFE Forecast

Fiscal year	Revenue ($m)	FCFE ($m)
FY2027	371,659.7	26,635.0
FY2028	412,542.2	39,032.5
FY2029	453,796.5	51,409.6
FY2030	494,638.2	63,545.0
FY2031	534,209.2	75,129.9

Revenue and FCFE increase across the projection period, but the model assumes substantial ongoing capital investment.

6.4 FCFE Valuation Results

Valuation component	Amount ($m)
Present value of explicit FCFE	185,148.6
Present value of terminal value	686,417.7
Equity value	871,566.2
Value per diluted share	**$116.94**

The FCFE valuation is lower than the FCFF valuation.

The difference should not be interpreted as an error in arithmetic. The models use different cash-flow definitions and assumptions and have not been fully reconciled economically.

7. Comparable-Company Valuation

To complement the intrinsic valuation models, I conducted a relative valuation analysis using Oracle Corporation and Adobe Inc.

Oracle was selected because of its enterprise software and cloud infrastructure exposure.

Adobe was selected because of its subscription-based software model and recurring revenue characteristics.

Neither company is a perfect comparable for Microsoft. Microsoft differs in business diversification, cloud scale, profitability, growth opportunities, and capital intensity.

7.1 P/E Multiple Comparison

Company	Reference price	EPS	P/E
Microsoft (MSFT)	$501.02	$17.95	27.91x
Oracle (ORCL)	$141.32	$5.83	24.24x
Adobe (ADBE)	$286.08	$16.70	17.13x

The peer median P/E multiple is approximately **20.69x**.

Applying this multiple to Microsoft's reference EPS of $17.95 produces a median-implied valuation of **$371.30 per share**.

The peer-implied valuation range is **$307.49–$435.11 per share**.

These calculations reproduce the results from `comps.py`. The historical price inputs and EPS definitions still require independent verification to establish consistent valuation-date conventions.

7.2 Leave-One-Out Sensitivity

Excluded company	Remaining implied MSFT price	Change from median
Oracle	$307.49	−$63.81
Adobe	$435.11	+$63.81

The substantial changes demonstrate that the estimate is sensitive to peer selection.

Because the analysis includes only two peers, the median multiple should be treated as a cross-check rather than a precise measure of fair value.

7.3 Interpretation

The peer median-implied value of $371.30 is higher than both intrinsic valuation estimates.

This may reflect differences in forecast assumptions, cash-flow conversion, growth expectations, and the limitations of applying simplified valuation methods to Microsoft's business.

However, the peer-implied valuation remains below the reference share price of $501.02.

The comparable-company analysis therefore provides additional support for maintaining a cautious investment recommendation.

8. Sensitivity Analysis and Downside Testing

Sensitivity analysis was conducted to determine how changes in operating and valuation assumptions affect estimated intrinsic value.

8.1 FCFF Discount Rate and Terminal Growth Sensitivity

The FCFF discounted cash flow model evaluates how changes in the weighted average cost of capital (WACC) and perpetual terminal growth rate affect Microsoft's estimated intrinsic value.

The analysis uses the revised FY2026 starting FCFF of $69,446.2199 million.

Sensitivity Grid — Estimated Value per Diluted Share

WACC	Terminal Growth 2%	Terminal Growth 3%	Terminal Growth 4%
9%	$204.33	$230.77	$267.78
10% (Base)	$178.59	**$197.71**	$223.20
11%	$158.60	$172.94	$191.38

Key Findings

The base-case valuation is $197.71 per diluted share, using a 10% WACC and 3% terminal growth rate.

The lowest valuation in the tested sensitivity grid is $158.60 per share, assuming an 11% WACC and 2% terminal growth rate.

The highest valuation is $267.78 per share, assuming a 9% WACC and 4% terminal growth rate.

The valuation range demonstrates that Microsoft's estimated intrinsic value is highly sensitive to discount-rate and terminal-growth assumptions.

Even the highest valuation in this grid remains below the September 1, 2026 reference price of $501.02 used in the model.

Investment Interpretation

The sensitivity analysis reinforces the watch/defer recommendation. Under all nine tested combinations of WACC and terminal growth, the estimated intrinsic value remains below the reference market price.

However, the results depend on the model's underlying FCFF forecast, financing assumptions, and terminal-value methodology. They should not be interpreted as a complete range of all possible outcomes.

8.2 Revenue Growth Sensitivity

The pro-forma model tests changes to the FY2027–FY2031 revenue growth path.

The base-case growth assumptions are 12%, 11%, 10%, 9%, and 8%.

A higher-growth scenario adds two percentage points to each forecast year.

Under the previously tested higher-growth scenario, estimated FCFE value increased from $116.94 to approximately **$144.44 per share**.

This demonstrates the importance of sustained revenue expansion.

8.3 Gross Margin Sensitivity

The base gross-margin assumption is 68%.

Under a tested 70% gross-margin scenario, the model produced an estimated value of approximately **$131.18 per share**.

This shows that improvements in operating profitability can increase estimated equity value.

8.4 Capital Expenditure Sensitivity

The interactive workbench also allows changes to the capital spending forecast.

Higher capital expenditures generally reduce cash available to equity investors unless the additional investment produces offsetting future operating benefits.

This is a particularly important risk for Microsoft because of the scale of AI infrastructure investment.

8.5 Sensitivity Conclusion

The tested scenarios demonstrate that Microsoft's valuation is sensitive to revenue growth, operating profitability, capital spending, and discounting assumptions.

The results reinforce the need for a margin of safety rather than relying on a single point estimate.

9. FCFF and FCFE Valuation Reconciliation

The FCFF and FCFE models produce materially different equity values.

9.1 Annual Cash Flow Comparison

Fiscal year	FCFF ($m)	FCFE ($m)	Difference ($m)
FY2027	79,863.2	26,635.0	53,228.1
FY2028	90,245.4	39,032.5	51,212.8
FY2029	100,172.4	51,409.6	48,762.8
FY2030	109,187.9	63,545.0	45,642.8
FY2031	116,831.0	75,129.9	41,701.1

These figures compare two distinct cash-flow measures, not economically interchangeable quantities.

9.2 Valuation Component Comparison

Component	FCFF DCF ($m)	FCFE pro-forma ($m)
PV of explicit cash flows	369,566.4	185,148.6
PV of terminal value	1,067,416.5	686,417.7
Equity value	1,473,531.9	871,566.2
Value per diluted share	$197.71	$116.94

The valuation difference is approximately **$80.77 per share**.

9.3 Reasons for the Difference

The two models differ in several ways.

Cash-flow definition: FCFF measures cash flow available to providers of debt and equity capital, while FCFE measures cash flow available to equity holders.

Forecast construction: The FCFF model applies growth assumptions to starting FCFF. The FCFE model derives cash flow from a more detailed operating and balance-sheet forecast.

Capital expenditures: The FCFE model explicitly forecasts substantial capital spending throughout FY2027–FY2031.

Terminal value: The models use different final-year cash-flow levels, producing materially different terminal values.

Enterprise-to-equity bridge: The FCFF model separately adjusts enterprise value for cash and debt. The FCFE model values equity directly.

A complete economic reconciliation would require aligning operating forecasts, financing flows, discount rates, and nonoperating adjustments.

The current comparison is therefore descriptive rather than a proof that both models are internally equivalent.

Supporting file: `project1_valuation_reconciliation.md`.

10. Interactive Valuation Workbench

I developed a Streamlit application to allow users to explore Microsoft's valuation assumptions interactively.

The application provides controls for:

- Revenue growth adjustments.
- Gross-margin assumptions.
- Capital expenditure multipliers.
- Cost of equity.
- FCFE terminal growth.
- WACC.
- FCFF terminal growth.
- Minimum cash requirements.

The workbench displays the FCFE and FCFF valuation estimates, projected financial statements, and validation warnings.

Base-Case Dashboard Outputs

Metric	Result
FCFE pro-forma / share	$116.94

Metric	Result
FCFF DCF / share	$197.71
FY2031 revenue	$534,209 million
FY2031 FCFE	$75,130 million

The application was restarted after the FCFF model update, and the revised FCFF result was confirmed in the dashboard.

The two valuation models remain separate. Revenue, margin, and capital expenditure controls primarily affect the FCFE model, while WACC and FCFF terminal growth affect the independent FCFF DCF.

The dashboard does not represent a fully reconciled single valuation model.

Supporting file: `app.py`.

11. Model Validation and Quality Control

The financial modeling process included numerical, accounting, and scenario checks.

11.1 Reproducibility

The FCFF model was executed using `dcf.py`.

The FCFE pro-forma was executed using `msft_proforma.py`.

The comparable-company analysis was executed using `comps.py`.

The reconciliation results were reproduced using `project1_reconciliation.py`.

11.2 Accounting Validation

The pro-forma model includes checks for balance-sheet consistency and other financial constraints.

The base-case model passed its reported balance-sheet checks.

11.3 Failure and Recovery Test

The Streamlit workbench was tested using an elevated minimum cash requirement.

When the minimum cash requirement was increased to $100,000 million, the application displayed a validation warning.

Restoring the minimum cash requirement to zero returned the model to its base-case configuration.

This demonstrates that the application can identify and communicate at least one invalid or constrained scenario.

11.4 Historical Input Review

The original DCF used a starting FCFF of $69,408.2199 million. After reviewing Microsoft's FY2026 Form 10-K, the operating cash flow and capital expenditure inputs were corrected, increasing starting FCFF to

$69,446.2199 million.

This correction increased the estimated FCFF valuation from approximately $197.60 to $197.71 per diluted share. The revised historical inputs and after-tax interest calculation are documented in `2026-09-10_MSFT_DCF_Inputs.md`.

Supporting files: `project1_validation.md` and `project1_source_verification.md`.

12. Investment Decision and Margin of Safety

The September 1, 2026 reference share price used in the models is $501.02.

Valuation Comparison

Method	Estimated value	Difference from reference price
FCFE pro-forma	$116.94	−$384.08
FCFF DCF	$197.71	−$303.31
Peer median P/E	$371.30	−$129.72
Reference share price	$501.02	—

The estimates do not establish a margin of safety at the reference price.

The comparable-company analysis produces the highest estimated value, but even its upper peer-implied estimate of $435.11 is below the reference price.

The intrinsic models are also subject to meaningful forecast and methodology uncertainty.

Given these findings, I would not recommend initiating a Microsoft position based solely on the current evidence.

Conditions That Could Change the Recommendation

I would reconsider the recommendation if:

1. Microsoft demonstrated sustained improvement in free-cash-flow conversion from AI and cloud investments.
2. Revenue growth and operating margins exceeded conservative expectations without requiring proportionately higher capital expenditures.
3. A revised and validated intrinsic valuation indicated a sufficient margin of safety relative to the market price.
4. Additional evidence supported higher sustainable returns on infrastructure investment.

The thesis would weaken further if cloud growth slowed materially, capital expenditures remained elevated without sufficient cash-flow improvement, or operating margins deteriorated.

13. Falsification and Investment Risks

A central falsification question is:

What evidence would demonstrate that Microsoft's AI infrastructure investments are generating sufficient incremental free cash flow to justify a higher intrinsic valuation and change the watch/defer recommendation?

This question connects the investment thesis to measurable financial outcomes rather than relying only on a favorable narrative.

Relevant indicators include Azure revenue growth, cloud gross margins, capital expenditures, depreciation, free-cash-flow conversion, and returns on invested capital.

This is an analyst-formulated question unless separately confirmed as the actual wording of a partner's question.

14. AI Use and Analyst Responsibility

AI assistance was used to support research organization, Python troubleshooting, model documentation, draft writing, and the preparation of financial analysis materials.

The analyst remains responsible for verifying primary-source financial information, selecting and defending model assumptions, interpreting results, and making the investment recommendation.

AI-generated explanations and calculations were checked against model outputs where possible.

Remaining unverified assumptions and methodological limitations are disclosed rather than presented as established facts.

15. Final Investment Conclusion

Microsoft has substantial competitive strengths and meaningful opportunities in cloud computing, enterprise software, and artificial intelligence.

However, my investment decision must consider valuation, capital intensity, forecast uncertainty, and downside risk.

The FCFF DCF estimates approximately $197.71 per share, the FCFE pro-forma estimates approximately $116.94 per share, and the comparable-company median P/E analysis implies approximately $371.30 per share.

All three estimates are below the $501.02 reference market price used in the analysis.

The results support maintaining a **WATCH / DEFER** recommendation rather than initiating a position.

The historical FY2026 cash-flow inputs and September 1, 2026 Microsoft reference price have been checked against their respective sources. However, the comparable-company price and EPS conventions require additional verification, and the differences between the FCFF and FCFE models remain an important methodological limitation. The valuation estimates should therefore be interpreted as scenario-based conclusions rather than precise predictions of future share prices.

16. Supporting Files and Reproducibility

The following files support this research:

File	Purpose
Project_1_Edition_A_MSFT.md	Original investment thesis
Project_1_Edition_B_MSFT.md	Updated research and investment analysis
dcf.py	FCFF DCF valuation
msft_proforma.py	Financial forecast and FCFE valuation
comps.py	Comparable-company valuation
project1_reconciliation.py	FCFF versus FCFE comparison
project1_valuation_reconciliation.md	Reconciliation documentation
project1_source_verification.md	Historical input review
project1_validation.md	Model validation and AI-use documentation
app.py	Interactive Streamlit valuation workbench

Reproduction Commands

From the TICKER-research directory, with the Python environment activated:

```
python dcf.py
python msft_proforma.py
python comps.py
python project1_reconciliation.py
python -m streamlit run app.py
```

The Streamlit application runs locally at http://localhost:8501.

17. Sources

Primary source

Microsoft Corporation. *FY2026 Form 10-K*, fiscal year ended June 30, 2026.

https://www.sec.gov/Archives/edgar/data/789019/000119312526323660/msft-20260630.htm

Additional research and model records

- MSFT-research/2026-09-10_Microsoft_Company_Research_Report.md
- 2026-09-10_MSFT_DCF_Inputs.md
- sources.md
- dcf.py
- msft_proforma.py
- comps.py

The Microsoft reference closing price of $501.02 on September 1, 2026, has been checked against Yahoo Finance historical data. The comparable-

company share prices and EPS inputs still require independent source and date verification. Analyst forecast assumptions remain subject to sensitivity analysis and investment judgment.

Edition B decision: WATCH / DEFER

Research status: Working draft; subject to final financial input verification, valuation consistency review, and submission formatting.